505-C

L-P-505c
February 5, 1974
Superseding
Fed. Spec. L-P-505b
September 24, 1964

L-P-505C

FEDERAL SPECIFICATION

PLASTIC PANELS, CORRUGATED, TRANSLUCENT, GLAZING

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers shatter-resistant, rigid, translucent, corrugated, glass fiber reinforced plastic (polyester, acrylic, or combination thereof) glazing panels for use on buildings where the exterior siding is corrugated (see 6.3).

1.2 Classification. The corrugated plastic glazing panels shall be of the following types, grades, and styles, as specified (see 6.2).

Type I	-	General purpose
Type II	-	Flame retardant (see 6.4)
Grade A	-	Light weight (5 oz. per sq. ft.)
Grade B	-	Standard weight (6 oz. per sq. ft.)
Grade C	-	Heavy weight (8 oz. per sq. ft.)
Style 1	-	2.67 Corrugated (Corrugation pitch 2.670 inches, depth 0.812 inch.)
Style 2	-	1-1/4 Corrugated (Corrugation pitch 1.250 inches, depth 0.250 inch.)
Style 3	-	2.67 Box Rib (Corrugation pitch 2.670 inches, depth 0.562 inch.)
Style 4	-	4.2 Corrugated (Corrugation pitch 4.200 inches, depth 1.062 inches.)
Style 5	-	2-1/2 Corrugated (Corrugation pitch 2.670 inches, depth 0.500 inch.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:10Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple Wall.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specificaton and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Dallas, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activites may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:10Z
Check the source to verify that this is the current version before use.

American Society for Testing and Materials (ASTM)

D 635 - Method of Test for Flammability of Self-Supporting Plastics.
D 757 - Method of Test for Flammability of Plastics, Self-Extinguishing Type.
D 1494 - Method of Test for Diffuse Light Transmission Factor of Reinforced Plastics Panels.
D 1499 - Recommended Practice for Operating Light-and Water-Exposure Apparatus(Carbon-Arc type) for Exposure of Plastics.
D 1502 - Method of Test for Transverse Load of Corrugated Reinforced Plastic Panels.
D 1602 - Method of Test for Bearing Load of Corrugated Plastics Panels.
D 2584 - Method of Test for Ignition Loss of Cured Reinforced Resins.
G 23 - Recommended Practice of Operating Light-and Water-Exposure apparatus (Carbon-Arc Type) for Exposure of Nonmetallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. The plastic glazing panels shall be composed of a glass fiber reinforced polyester, acrylic, or combination thereof, resin. The resin shall be light-stabilized and shall contain minor amounts of pigments, dyes, and opacifying fillers. The glass reinforcement shall be of the chopped-fiber, random strand type or continous strand, glass mat type. The glazing panels shall be rigid, corrugated, translucent, non-porous, and shatter-resistant; and be capable of being sawed and nailed in place without cracking, splitting, breaking, or damaging the glazing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:10Z
Check the source to verify that this is the current version before use.

3.1.1 Types. Type I panels shall have a rate of burning of less than 2.0 inches per minute and type II panels shall have a rate of burning of less than 0.35 inch per minute when tested in accordance with 4.4.1 (see 6.4).

3.1.2 Grades. Grades A, B, and C panels shall have the weights specified in 1.2 with a tolerance of plus or minus 10 percent when tested in accordance with 4.3.1.

3.1.3 Styles. Each style shall have the pitch and depth dimensions specified in 1.2 with a pitch tolerance of plus 0.010 and minus 0.015 inch and a depth tolerance of plus or minus 0.100 inch when tested in accordance with 4.3.2.1 and 4.3.2.2.

3.2 Sizes. Panels shall be furnished in the lengths and widths as specified (see 6.2 and 6.5). The tolerance on the specified lengths and widths shall be ± 1/4 inch when measured in accordance with 4.3.2.3.

3.3 Color. The panels shall be light green in color, unless otherwise specified (see 6.2).

3.4 Finish. The panels shall be furnished with a smooth finish on both sides, unless otherwise specified (see 6.2 and 6.6).

3.5 Panel squareness. Panels 27-1/2 inches or less in width shall be within 1/16 inch of square; and those wider than 27-1/2 inches shall be within 1/8 inch of square, when measured in accordance with 4.3.3.

3.6 Glass fiber content. The glass content shall represent not less than 20 percent by weight of the formed panel when tested in accordance with 4.4.2.

3.7 Light transmission (light green panels only). The minimum light transmission shall be 75.0 percent for grade A panels, 70.0 percent for grade B panels, and 65.0 percent for grade C panels when tested in accordance with 4.4.3 (see 3.3).

3.8 Heat distortion. The panels shall have an average heat distortion no greater than 0.50 inch after 48 hours at 180°F when tested in accordance with 4.4.4.

3.9 Transverse load. When tested in accordance with 4.4.5, the panels shall support the minimum transverse loads shown in Table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:10Z
Check the source to verify that this is the current version before use.

TABLE I. Transverse Loads (Minimum)[1]

Style	Type I			Type II		
	Grade A	Grade B	Grade C	Grade A	Grade B	Grade C
	(lb. per ft. of width)			(lb. per ft. of width)		
1	200	300	450	110	175	250
2	90	120	200	70	90	150
3	-	185	300	-	160	250
4	180	280	400	135	210	300
5	135	210	300	110	175	250

1 These values are for quality control purposes only, and are not suitable for direct use in design, engineering, or architectural calculations.

3.10 Bearing load. When tested in accordance with 4.4.6, the panels shall have the bearing loads shown in Table II.

TABLE II. Bearing Loads

Grade	Average Load	Minimum Specimen Load
	lb	lb
A	90	70
B	145	115
C	225	175

3.11 Accelerated weathering resistance.

3.11.1 Type I. When tested in accordance with 4.4.7, type I panels shall show no evidence of brittleness, cracking, crazing, or delamination, and show no more than slight to moderate visual changes in color, gloss, and surface fiber prominence.

3.11.2 Type II. When tested in accordance with 4.4.7, type II panels shall show no evidence of brittleness, cracking, crazing, or delamination, and show no more than moderate to extensive visual changes in color, gloss, and surface fiber prominence.

3.12 Workmanship. The plastic glazing panels shall be uniform in appearance, shall contain no resin-starved areas or air cavities larger than 1/32 inch in diameter, and shall be free from warpage, cracks, scratches, and other defects that affect the appearance or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:10Z
Check the source to verify that this is the current version before use.

facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Quality conformance inspection shall include all the examinations and tests of this specification and shall be performed on every lot of plastic glazing panels offered for delivery.

4.2.1 Lot. All plastic glazing panels of the same type, grade and style offered for delivery at the same time under a contract or order shall be considered a lot for purposes of inspection.

4.2.2 Sampling for examination. A random sample of panels shall be selected from each lot in accordance with MIL-STD-105, using inspection level I and an acceptable quality level of 2.5.

4.2.3 Sampling for tests. A random sample of panels shall be selected from each lot in accordance with MIL-STD-105, using inspection level S-1 and an acceptable quality level of 6.5, except that the minimum sample size shall be not less than 3.

4.3 Examination. Sample panels selected in accordance with 4.2.2 shall be examined for incorrect weight, dimensions, color, finish, and panel squareness, and for visual defects in workmanship. Any sample panel having one or more of the defects listed shall be considered a defective unit.

4.3.1 Weight. Panels shall be weighed individually on a beam scale with a capacity of 50 pounds and an accuracy within one percent. The scale shall be checked for accuracy before the test is performed and adjusted if necessary. The weight in ounces per square foot of each formed panel shall be calculated.

4.3.2 Dimensions. The panel shall be laid on a flat, smooth surface and measured for the following dimensions:

4.3.2.1 Pitch. The pitch is the average distance from the crest of one corrugation to the crest of an adjacent corrugation. The pitch shall be determined by placing a metal straightedge crosswise on the panel so that it touches the crests and measuring the distance between the crests to the nearest 0.01 inch. Ten measurements shall be made on various parts of the panel and the results averaged.

4.3.2.2 Depth. The depth of the corrugation is the vertical distance between the plane of the crests and the upper side of the panel at the bottom of the valley. Ten depth measurements, five at each end, shall be made on the panel with a depth micrometer, measuring to the nearest 0.03 inch, and the results averaged.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:10Z
Check the source to verify that this is the current version before use.

4.3.2.3 Length and width. The panel shall be measured with a steel tape graduated in 1/32-inch increments. The length shall be measured on the two sides and the center to the nearest 1/32-inch, and the three measurements averaged. The width shall be measured on the projected width at each end and in the center to the nearest 1/32-inch, and the three measurements averaged.

4.3.3 Panel squareness. Panel squareness shall be determined on full length panels as offered for delivery. The examination shall be made on a flat table that has two rails mounted on it at 90 degrees to one another. The panel shall be placed on the table so that one edge of the panel touches one rail along its entire length and the corner of the panel is in the 90 degree angle between the rails. The amount of out-of-squareness shall be the widest gap between the second edge of the panel and the other rail, measured to the nearest 1/32-inch. The examination shall be repeated so that all four corners of each panel are measured for squareness.

4.4 Tests. Test specimens taken from sample panels selected in accordance with 4.2.3 shall be conditioned for not less than 40 hours at 23°C (73.4°F) plus or minus 2.0°C (3.6°F) and 50 plus or minus 5 percent relative humidity and tested at the same temperature and humidity, unless otherwise specified in the particular test method.

4.4.1 Rate of burning. The rate of burning shall be determined in accordance with ASTM D635, except that six specimens taken from different parts of the panel shall be tested and the results averaged. Two specimens shall be taken from crests, two from sides of valleys, and two from bottoms of valleys. When the specimens of type II panels do not burn to the gage mark when tested by ASTM D635, their rate of burning shall be determined in accordance with ASTM D757, except that the panels shall be tested regardless of the thickness of the test specimens.

4.4.2 Glass fiber content. A sample panel shall be divided into six equal parts and a representative test specimen taken from each part, the size to be one complete corrugation by one inch in length. Each specimen shall be tested for ignition loss in accordance with ASTM D2584, the percent glass content shall be calculated, and the results averaged.

4.4.3 Light transmission. Light transmission shall be determined in accordance with ASTM D1494.

4.4.4 Heat Distortion. Heat distortion shall be determined on three test specimens cut from a sample panel. Each specimen shall be 6 by 12 inches, and cut with the corrugations parallel to the 12-inch dimension. The specimens shall be tested in an oven capable of maintaining a temperature of 82.2°C

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:10Z
Check the source to verify that this is the current version before use.

(180°F) plus or minus 2°C (3.6°F) and of sufficient size to hold a test jig to support the specimens. The supports on the test jig shall have a 0.5-inch radius, shall be oriented to provide a 10-inch span and accomodate a specimen six inches wide, and shall be provided with a means of measuring deflection to the nearest 0.01 inch. The test jig shall be placed in the oven and the oven preheated to the above test temperature. Each specimen in turn shall be placed on the supports of the test jig which had been adjusted to provide a 10-inch span, and the deflection device shall be positioned to measure the specimen deflection at mid-span and in the center of the specimen width. The oven temperature shall be brought back to the test temperature and the deflectometer read. After 48 hours, the deflectometer shall be read again and the difference in the readings shall be recorded as the deflection or sag. The average of the deflections recorded for the three specimens shall be reported as the heat distortion of the panel.

4.4.5 Transverse load. Transverse load shall be determined in accordance with ASTM D1502.

4.4.6 Bearing load. Bearing load shall be determined in accordance with ASTM D1602.

4.4.7 Accelerated weathering resistance. Accelerated weathering resistance shall be determined in accordance with ASTM G23 and ASTM D1499. Three test specimens shall be cut from each sample panel and exposed for 1000 plus or minus 50 hours in a type E, F, or G single open-flame sunshine carbon-arc lamp apparatus. The spray water used in the test shall be distilled and/or deionized water. Following the exposure period, the specimens shall be removed from the apparatus and visually examined.

4.5 Certificate of compliance. Unless otherwise specified (see 6.2), a certificate of compliance shall be accepted as proof that the requirements of the accelerated weathering resistance test in 4.4.7 have been met.

4.6 Inspection of preparation for delivery. The packing of the plastic glazing panels and marking of the containers for shipment shall be inspected to determine conformance with the applicable requirements of Section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. The panels shall be packed in boxes conforming to PPP-B-636 (V3c) or PPP-B-640 (class 2). Flat steel strapping shall be type I, Finish B.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:10Z
Check the source to verify that this is the current version before use.

5.1.2 Level B. The panels shall be packed as specified for level A, except that the boxes shall be domestic class, and flat steel strapping need not be Finish B.

5.1.3 Level C. The panels shall be packed in a manner that will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.2 Marking.

5.2.1 Civil agencies. In addition to any marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to any marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The fiberglass reinforced plastic panels covered by this specification are intended for use as glazing in light transmission and decorative applications in construction where the exterior siding is corrugated. The principal differences between the two types of panels are in their resistance to burning and weathering. Type I has better weathering properties than type II, but type II has a slower rate of burning obtained at some sacrifice in resistance to weathering. Type II may cost more than type I.

6.2 Ordering Data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.

(b) Type, grade, and style of panels required (see 1.2 and 6.1).

(c) Sizes of panels required (see 3.2 and 6.5).

(d) Color required, if other than light green (see 3.3).

(e) Finish required if other than a smooth finish on both sides (see 3.4 and 6.6).

(f) If a certificate of compliance for accelerated weathering resistance is not acceptable (see 4.5).

(g) Level of packing required (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:10Z
Check the source to verify that this is the current version before use.

6.2.2 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Armed Services Procurement Regulation, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR 7-104.9(n) are not invoked, the data specified below shall be delivered in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraph.

Paragraph	Data Requirement	Applicable DD 1664
4.5	Certificate of Compliance	DI-E-2121

(Copies of Data Item Descriptions required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 Installation. Installation guidelines for the panels are provided in a brochure titled "Recommended Installation Practice for Fiberglass Reinforced Panels" obtainable from the Society of the Plastics Industry, Inc., 250 Park Avenue, New York, New York 10017.

6.4 Flame retardant. The term "flame retardant" and the rates of burning used to distinguish type II panels from type I panels are based on small scale laboratory screening tests to compare relative flammability. The tests are used solely to establish relative burning characteristics, and the term "flame retardant" should not be considered a fire-hazard classification, but only as an aid in distinguishing between the two types of panels.

6.5 Sizes available. The most common sizes of panels currently available are 6 to 16 feet in length and 26 to 48 inches in width.

6.6 Finishes available. The panels are commonly available in the following three finishes:

(a) Smooth finish on both sides.

(b) Textured finish on both sides.

(c) Smooth finish on one side and textured finish on the other side.

MILITARY CUSTODIANS:
ARMY-MR
NAVY-YD
AIR FORCE-11

User Activities:
ARMY - EL, MI

Preparing Activity:
NAVY - YD

Civil Agency Coordinating Activity:
GSA-FSS

Project No. 9330-0571

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 15 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:10Z
Check the source to verify that this is the current version before use.